

Question	Marking Guidance	Mark	Comments
01.1	1. Laboratory-raised female (guppies) might not react/behave/choose in the same way (as wild guppies); 2. (Transparent) barrier might not allow for normal (courtship) behaviour/interaction; 3. Do not know if (guppies) have been used in previous experiments; 4. 10 minutes might not be long enough for females to make a (final) choice OR Not enough time for females to make a (final) choice;	3 max	Ignore answers relating to sample size 1. Accept laboratory-raised female (guppies) might not be representative of wild females 2. Accept choice might involve chemical/mechanical signals/interaction 2. Accept colour might not be the only thing females are attracted to 4. Accept descriptions of a choice eg 'show attraction'
01.2	1. (Females with large brains) will mate with males bright in colour; 2. Their (male) offspring would be (more likely to be) bright in colour; 3. (Bright in colour male) offspring could attract larger brained females; 4. The population/offspring could (evolve to) have larger brains; 5. The population/offspring are better at identifying/avoiding predators;	3 max	Accept answers that include references to alleles 4. and 5. Ignore answers relating to females only

01.3	1. Not geographically isolated; 2. (Leading to) reproductive isolation OR Gene pools kept separate; 3. Changes in allele <u>frequencies</u>; 4. Cannot breed/mate to produce fertile offspring;	3 max	1. Accept are in the same area 2. Accept large brained females will only mate with males bright in colour <u>and</u> small brained females will only mate with males dull in colour 3. Reject gene frequencies 4. Reject inbreeding
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